Material-based Hertz sphere-wall impact

LAMMPS dem03 hertz/material | Compression, force, energy and wall impulse | 28 October 2026

PASS	4 of 4 runs pass the official 1% peak-energy check at 21.38 us.

Compression and force peaks

Run	Overlap (um)	Err. (%)	Force (N)	Err. (%)	Energy err. (%)	Official
CPU dt	22.680647	0.0000133	384.052517	0.0000199	0.0000332	PASS
GPU dt	22.680647	0.0000133	384.052517	0.0000199	0.0000332	PASS
CPU dt/2	22.680644	0.0000016	384.052449	0.0000024	0.0000040	PASS
GPU dt/2	22.680644	0.0000016	384.052449	0.0000024	0.0000040	PASS

Independent targets: overlap = 22.680643558 um; force = 384.052440292 N. Peaks use every native step. The additional overlap/force 1% limits are diagnostics; the official checker asserts the sampled peak-energy relation.

Direct material inputs and fixed-plane mapping

Quantity	Official input / FunDEM configuration
One sphere / plane	Diameter 0.005 m; density 7000 kg/m^3; initial center z = 0.002550 m; vz = -3.9 m/s; zero spin. Fixed plane z = 0.
Normal law	hertz/material E en nu; damping coeff_restitution. E = 100 GPa; nu = 0.25; en = 1. F = (4/3) E* sqrt(R) delta^(3/2).
Finite elastic wall	Both sphere and wall use the official E and nu directly: E* = E / [2 (1 - nu^2)] = 53.3333333 GPa. Fixed wall motion does not imply infinite modulus.
Damping / other forces	Nonlinear stiffness; manual damping override off; restitution 1 gives zero automatic damping. All friction, gravity, global damping, spin and torque = 0.
Time / full output	dt = 10 / 5 ns; 3038 / 6076 steps; same 30.38 us endpoint; native frames and CSV every step, including t = 0.

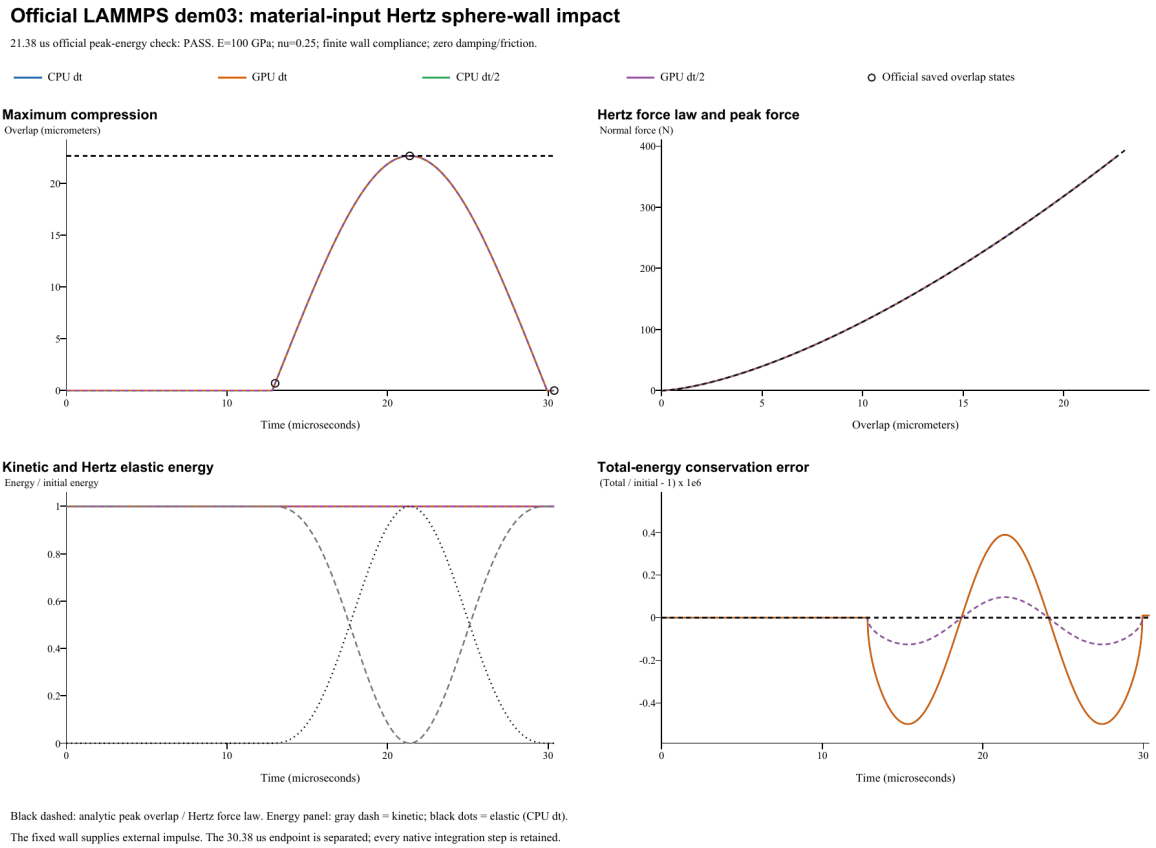
Official energy criterion at 21.38 us

E0 = (1/2) m V^2 = 0.003484222602 J; V = 3.9 m/s; R = 0.0025 m.
U = (2/5) K delta^(5/2) = 0.4 F delta; K = (4/3) E* sqrt(R).
The official checker compares E0 with 0.4 abs(F1) delta at step 2138 using 1% tolerance. It does not search for separate maxima or add residual kinetic energy at this near-peak instant.

Reference provenance

Official YAML and saved states | Analytic checker. Commit f9720ce2d176d615495d1d6c31349b1f71be1c73; YAML blob 2b1e846a6377f3218c0bc3756e3c430738d12a33. LAMMPS was not rerun locally. YAML epsilon 1e-10 is not a universal cross-code tolerance.

Energy histories and retained evidence



Whole-event energy and wall impulse

Run	Max. energy err. (%)	Final energy err. (%)	Rebound restitution
CPU dt	0.0000499	0.0000010	1.0000000051
GPU dt	0.0000499	0.0000010	1.0000000051
CPU dt/2	0.0000126	0.0000001	1.0000000003
GPU dt/2	0.0000126	0.0000001	1.0000000003

Maximum whole-event energy-error limit: a separate 1% diagnostic. E = sphere kinetic energy plus one wall-contact elastic energy. At the 30.38 us endpoint the sphere has separated; the rebound ratio is a valid free-flight diagnostic, not a separate official assertion.

CPU/GPU velocities differ by at most 4.44e-16 m/s; maximum force-history difference is 0 N. Halving dt changes maximum whole-event energy error to 0.251777 of its base-step value. The official fixed-time PE error includes near-peak residual kinetic energy and differs from the conservation error.

The particle's momentum is not conserved. The fixed wall's integrated contact impulse balances its momentum change; the support receives the opposite impulse. The fixed plane has no kinetic energy or boundary work.

Complete native results are retained

CPU dt: run.json | GPU dt: run.json | CPU dt/2: run.json | GPU dt/2: run.json

3039 frames per base-step run; 6077 per half-step run. All 18,232 native frames, contacts, projects, every-step CSVs, audits and backend logs are retained.

Run manifest | Analysis JSON | Checkpoint CSV | Contact audit | Full native archive ZIP

Execution and scope

Real SimulationSession runs use FunDEM 0.4.3 CPU/CUDA native libraries; no solver changes or fitting. GUI executable byte identity was not established. Scope: direct E/nu, finite wall compliance, Hertz peaks, energy and wall impulse. Dissipative, tangential, rotating and many-particle cases require other tests.